

PET Imaging in Alzheimer Disease in the Era of Antiamyloid Therapy in the United States: Clinical Utility, Quantification, and Policy Landscape

Ty Skyles^{*1}, Samantha M. Bouchal^{*1,2}, Anna Giarratana³, Jacob Wengler¹, Ian Hart¹, Erin Greig⁴, Harmanjeet Singh⁵, Steve S. Huang⁶, Felipe Martinez⁶, Ba Nguyen⁶, Clifford H. Shin⁶, Ming Yang⁶, Ephraim Parent⁷, W. Hudson Robb⁶, Ana M. Franceschi⁸, Brian Burkett⁹, Derek Johnson⁹, and Mary Ellen Koran⁶

¹Mayo Clinic Alix School of Medicine, Phoenix, Arizona; ²Mayo Clinic Medical Scientist Training Program, Rochester, Minnesota; ³Department of Radiology, University of Wisconsin, Madison, Wisconsin; ⁴Department of Neuroscience, Mayo Clinic, Jacksonville, Florida; ⁵Mahatma Gandhi Memorial Medical College, Jamshedpur, India; ⁶Department of Radiology, Mayo Clinic, Scottsdale, Arizona; ⁷Department of Radiology, Mayo Clinic, Jacksonville, Florida; ⁸Neuroradiology Division, Department of Radiology, Donald and Barbara Zucker School of Medicine at Hofstra/Northwell, Lenox Hill Hospital, New York, New York; and ⁹Department of Radiology, Mayo Clinic, Rochester, Minnesota

CE credit: For CE credit, you can access the test for this article, as well as additional JNMT CE tests, online at <https://www.snmlearningcenter.org>. You may make 3 attempts to pass the test and must answer 80% of the questions correctly to receive credit—number of credits awarded will be determined by the length of the article. Participants can access this activity through the SNMMI website (<http://www.snmlearningcenter.org>) through March 2029. Additional details such as the number of credits issued per article, expiration dates, financial disclosure information, VOICE transcripts, and the process to earn CE credit can also be found in the SNMMI Learning Center.

Alzheimer disease (AD) is increasingly diagnosed using molecular imaging biomarkers. PET imaging provides the opportunity to visualize amyloid and tau aggregates and in vivo neurodegenerative changes. These techniques provide exciting new avenues toward diagnosis, disease staging, and therapeutic monitoring of AD. **Methods:** This review details recent advances in amyloid PET, tau PET, and ¹⁸F-FDG PET as they relate to the diagnosis, staging, and treatment of AD. The increasing roles of PET in the biologically based diagnosis of AD and antiamyloid immunotherapy response monitoring are addressed. **Results:** Amyloid PET enables improved detection of amyloid- β plaques within the brain. Amyloid PET is increasingly vital for confirming AD diagnoses given the emergence of antiamyloid immunotherapies, which require biomarker-verified amyloid positivity to initiate treatment. Tau PET provides a direct measure of neurofibrillary tangle pathology and is useful for disease staging, the interpretation of atypical clinical presentations, and treatment decision-making. ¹⁸F-FDG PET plays a vital role in distinguishing AD from other dementia subtypes. Expanded reimbursement policies for amyloid and tau PET have increased accessibility to these modalities. Finally, quantitative methods facilitate interscan reproducibility and permit therapeutic monitoring. **Conclusion:** Molecular neuroimaging is poised to play a central role in the biologic definition, diagnosis, staging, and management of AD. Integrating amyloid, tau, and FDG PET with clinical assessments and fluid biomarkers provides earlier and more accurate diagnoses, opening the door to personalized treatment.

Keywords: neurology; radiopharmaceuticals; Alzheimer disease; amyloid PET; antiamyloid immunotherapy; tau PET

Received Dec. 8, 2025; revision accepted Jan. 16, 2026.
For correspondence, contact Mary Ellen Koran (koran.maryellen@mayo.edu).

^{*}Contributed equally to this work.

Published online Feb. 10, 2026.

COPYRIGHT © 2026 by the Society of Nuclear Medicine and Molecular Imaging.

J Nucl Med Technol 2026; 00:1–8

DOI: 10.2967/jnmt.125.271835

Alzheimer disease (AD) is the most common cause of dementia worldwide. With nearly 14 million Americans projected to be living with AD by 2060, this disease represents a major public health burden (1). Traditionally, AD was a clinical diagnosis confirmed at autopsy by the identification of hallmark neuropathologic features (i.e., amyloid plaques and neurofibrillary tangles) (2). However, even in specialized dementia centers, antemortem diagnostic accuracy is far from ideal. Studies comparing autopsy results with clinical diagnoses have reported misdiagnosis rates of 25%–30% in dementia patients (3). This rate may be higher in nonspecialty or primary care settings and in patients for whom non-AD dementias are on the differential diagnosis (3). Multiple cohort studies indicate that many AD cases include additional neuropathologic comorbidities (mixed dementia) at autopsy, which may also impact diagnostic accuracy (4–6).

PET has transformed the AD diagnostic landscape by allowing in vivo visualization of hallmark pathologic proteins. AD is characterized by extracellular deposition of amyloid- β (A β) plaques and intracellular accumulation of abnormally phosphorylated tau into neurofibrillary tangles (7). The amyloid cascade hypothesis posits that the deposition of A β in the brain triggers tau accumulation, leading to neurodegeneration and culminating in symptomatic AD (8,9). Each step of this pathologic pathway can be imaged using brain PET and radiopharmaceuticals. Amyloid PET can image A β plaques (10), tau PET can image neurofibrillary

tangles (10), and ^{18}F -FDG PET can visualize neuronal cell dysfunction and neurodegeneration (11).

With the emergence of anti-amyloid monoclonal antibodies in the early 2020s, $\text{A}\beta$ and tau PET imaging now play pivotal roles in determining whether patients are therapeutic candidates and in evaluating treatment response (12). Expanded coverage of $\text{A}\beta$ PET by the Centers for Medicare and Medicaid Services (CMS) in 2023 (13) has made amyloid PET imaging significantly more accessible clinically; as of 2024–2025, CMS also began reimbursing tau PET scans. In 2024, the Alzheimer Association Workgroup revised their recommendations for the diagnosis of AD, moving toward a biologic diagnosis based on biomarkers, including PET imaging (14). This review examines the evolving diagnostic and clinical landscape of AD, specifically with respect to the utility, methodologic challenges, reimbursement, and clinical impact of ^{18}F -FDG, amyloid, and tau PET.

CLINICAL UTILITY OF PET IMAGING

PET imaging has become increasingly important in the clinical evaluation of AD. ^{18}F -FDG PET, which leverages radioactive glucose labeling to visualize metabolic activity, has been used for decades to assess regional patterns of hypometabolism in neurodegenerative disorders (15,16). Amyloid PET emerged in the early 2000s, with several radiopharmaceuticals developed to enable *in vivo* detection of $\text{A}\beta$ plaques (17). More recently, the development of tau PET radiopharmaceuticals has expanded the options for *in vivo* visualization of AD neuropathologic changes. Tau PET maps the distribution of neurofibrillary tangles throughout the brain, providing a more specific correlate to disease stage and clinical severity than amyloid or ^{18}F -FDG PET (18). Together, these modalities allow for an integrated approach to understanding, diagnosing, and managing AD.

^{18}F -FDG PET

^{18}F -FDG PET, originally used for differentiating AD from frontotemporal lobar degeneration, is now commonly used for the assessment of atypical dementias and neurodegenerative disorders with mixed pathology (19). ^{18}F -FDG PET signals reflect neuronal and synaptic activity, as well as microglial activation states and astroglial glutamate transport (20,21). In AD, hypometabolism is observed in the posterior cingulate, precuneus, and temporoparietal cortices, with relative preservation of metabolism in the sensorimotor cortex (Fig. 1). This pattern often produces the appearance of a person wearing headphones, sometimes referred to as the “earmuff” or “headphone” sign (22,23).

^{18}F -FDG PET is particularly useful in differentiating AD from other neurodegenerative disorders. ^{18}F -FDG PET can differentiate AD metabolic patterns from those of dementia with Lewy bodies, which presents with occipital and posterior cortical hypometabolism and sparing of the posterior cingulate (24), as well as vascular dementia, which presents with patchy, asymmetric hypometabolism patterns matching

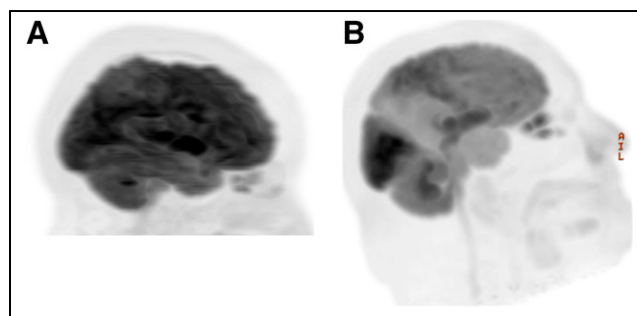


FIGURE 1. ^{18}F -FDG PET scans of patients without (A) and with (B) AD. (A) Maximum-intensity-projection image showing absence of gross atrophy or pathology. (B) Maximum-intensity-projection image showing characteristic hypometabolism in posterior cingulate, precuneus, and temporoparietal cortices, with relative preservation of metabolism in sensorimotor cortex. This pattern often produces appearance of person wearing headphones, sometimes referred to as “earmuff” or “headphone” sign.

infarct territories seen on structural imaging (25). More recently, ^{18}F -FDG PET has shown value in identifying emerging entities, such as limbic-predominant age-related TDP-43 encephalopathy, which can mimic AD clinically but displays characteristic anteromedial temporal hypometabolism (11). Although the medial anterior temporal lobes demonstrate lower physiologic ^{18}F -FDG uptake at baseline relative to the rest of the brain, this normal regional pattern can be misinterpreted as pathologic hypometabolism on visual assessment alone, leading to potential false-positive reads. Computer-assisted artificial intelligence–based quantitative approaches may be particularly valuable in this region by providing objective comparisons to normative reference datasets, allowing anterior medial temporal hypometabolism to be classified as abnormal only when it truly deviates from expected physiologic values.

Recently, academic centers have integrated artificial intelligence tools into their ^{18}F -FDG PET analysis pipelines. These tools compare a patient’s ^{18}F -FDG PET scans against those of large reference cohorts to estimate the likelihood of specific dementia subtypes (26,27), which can improve diagnostic precision, especially in cases with copathology, such as concomitant Lewy body proteinopathy, TDP-43 proteinopathy, or vascular brain injury.

AMYLOID PET

Visualizing $\text{A}\beta$ Pathology

Amyloid PET is a powerful tool that provides direct visualization of fibrillar $\text{A}\beta$, the defining hallmark of AD (28). The development of $\text{A}\beta$ PET radiopharmaceuticals began with the ^{11}C -Pittsburgh compound B, which was followed by the more widely used ^{18}F -labeled agents (e.g., florbetapir, florbetaben, flutemetamol). These radiopharmaceuticals have enabled the *in vivo* detection of amyloid (Fig. 2). ^{11}C -Pittsburgh compound B is unlikely to receive Food and Drug Administration (FDA) approval, as it requires an on-site cyclotron for synthesis, limiting scalability. The first

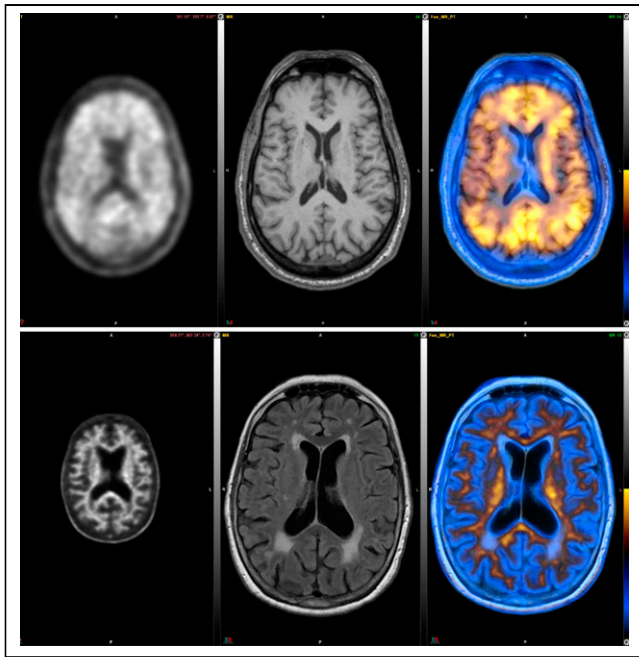


FIGURE 2. Positive vs. negative amyloid PET scans. Top panel shows, from left to right, positive florbetaben amyloid PET scan, corresponding T1 MRI, and fused PET/MRI. Radiopharmaceutical binds out to edge of neocortex, indicating presence of moderate to frequent abnormal neuritic A β plaques. Centiloid value of this scan (86) was concordant with visual read. Bottom panel shows, from left to right, negative amyloid PET scan, corresponding T2 fluid-attenuated inversion recovery MRI, and fused PET/MRI. Florbetaben binds only to cerebral white matter in a nonspecific pattern and does not indicate amyloid accumulation. This negative study effectively excludes AD pathology and reflects sparse to no neuritic A β plaques. Centiloid value of this scan (–26) was concordant with visual read. Of note, white matter binding of radiopharmaceutical is reduced in areas with fluid-attenuated inversion recovery hyperintensities.

^{18}F -labeled A β radiopharmaceutical, florbetapir, was approved by FDA in 2012 (29); the approval of florbetaben and flutemetamol followed quickly thereafter. A positive amyloid PET scan indicates the presence of moderate to frequent abnormal neuritic A β plaques, whereas a negative study indicates sparse to no neuritic A β plaques, effectively excluding AD pathology, even with a suggestive clinical presentation (Fig. 2) (30).

Clinically, amyloid PET assists in confirming AD pathology in diagnostically ambiguous cases. It has become particularly important in establishing patient eligibility for monoclonal anti-amyloid targeting therapies, as documented amyloid positivity is required before starting therapy (28). As of June 2025, the FDA granted expanded indications for all 3 FDA-approved radiopharmaceuticals to include quantitative amyloid plaque measurement and expanded indications for anti-amyloid therapy monitoring (31), on the basis of evidence showing that patients who had better therapeutic response on amyloid PET also experience the most clinical benefit (32).

Two key advantages of amyloid PET over fluid biomarkers of A β , such as cerebrospinal fluid (CSF) (A β 42/40

ratio) and plasma (i.e., phosphorylated tau 217) immunoassays, are the spatial resolution that PET affords (Fig. 2) and that amyloid PET can track treatment response. Fluid biomarkers are useful for the early detection of AD but currently lack spatial information (33). Additionally, plasma biomarkers do not yet track well with amyloid clearance after treatment (34), and patients may be averse to invasive short-interval repeat lumbar punctures for CSF. Finally, plasma phosphorylated tau 217 levels in the “intermediate” range (0.186–0.324 pg/mL) fall between assay cutoffs, often triggering further evaluation by amyloid PET or lumbar puncture for confirmation (35).

Reimbursement Challenges

The FDA determines the safety and efficacy of radiopharmaceuticals or scans, whereas CMS decides whether these services will be reimbursed for patients with Medicare or Medicaid. FDA approval does not guarantee CMS coverage, and CMS coverage decisions for Medicare and Medicaid beneficiaries often serve as a benchmark for private insurer coverage decisions. This separation of regulatory and reimbursement pathways can slow clinical adoption of novel imaging biomarkers and therapies (36).

When florbetapir received FDA approval in 2012 (Fig. 3), CMS authorized florbetapir reimbursement only for patients in an approved clinical research study, limiting its availability and use. One of the trials conducted under the Coverage with Evidence Development policy for florbetapir was the iDEAS trial, which imaged 18,295 patients diagnosed with mild cognitive impairment or dementia of uncertain cause. In 2019, iDEAS trial data demonstrated that amyloid PET changed clinical management in over 60% of cases and diagnoses (from AD to non-AD dementia or vice versa) in over 30% of cases (37). However, CMS maintained the Coverage with Evidence Development restriction until 2023, citing insufficient evidence that imaging improved patient outcomes (38).

In July 2023, lecanemab received traditional FDA approval and was the first anti-amyloid therapy to gain CMS reimbursement (39,40). However, amyloid PET remained uncovered at this time, forcing many patients to undergo lumbar puncture for CSF confirmation of amyloid pathology. This inconsistency was resolved in October 2023 when CMS lifted the Coverage with Evidence Development policy, allowing for multiple amyloid PET scans to be reimbursed (38). The long-term impact of amyloid PET imaging on patient-centered outcomes and cost-effectiveness is being investigated (41).

ANTI-AMYLOID THERAPIES

For decades, the only available therapies for AD targeted symptoms without altering the underlying disease course. The introduction of monoclonal antibodies targeted against A β represents the first effective disease-modifying approach. Aducanumab was the first FDA-approved anti-amyloid agent in 2021 (Fig. 3), but uncertain clinical benefit, safety

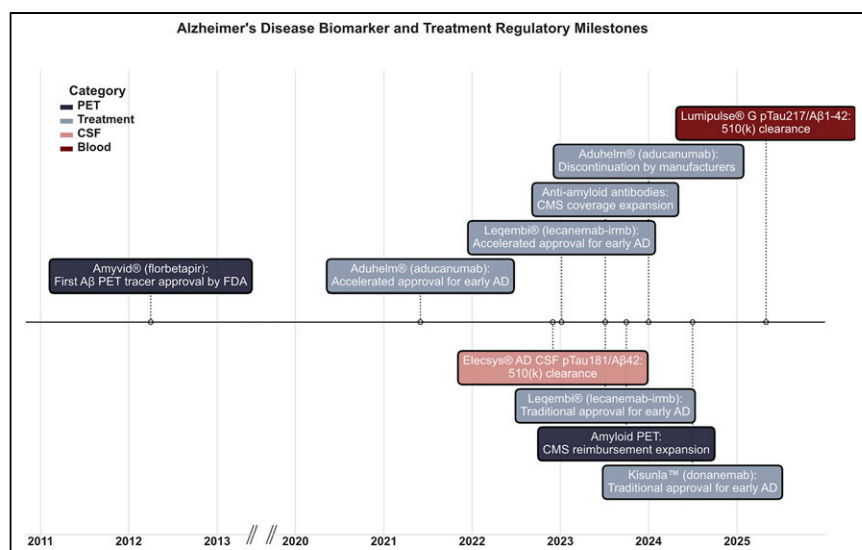


FIGURE 3. Regulatory timeline for AD biomarkers and therapeutics. Timeline highlighting key U.S. regulatory milestones for PET biomarkers, CSF and blood assays, and anti-amyloid therapies from 2011 to 2025. “Accelerated approval” and “standard approval” refer to FDA approval.

concerns, and the lack of CMS coverage limited its clinical adoption (42). In 2023, lecanemab became the first anti-amyloid therapy to achieve both traditional FDA approval and CMS reimbursement (39), followed by donanemab in 2024 (43). Both therapies are now in clinical use, with lecanemab administered biweekly and donanemab given monthly. Both have shown modest slowing of cognitive decline but carry risks of amyloid-related imaging abnormalities (ARIA) and require MRI monitoring (44,45).

Clinical Impact and Safety of Therapies

Lecanemab and donanemab have been shown to slow, not halt, the clinical progression of AD (46,47). With amyloid PET imaging, these therapies can significantly reduce cortical Aβ burden over the treatment course of 18 mo (47,48). Crucially, however, anti-amyloid antibodies carry the risk of ARIA, for which the FDA has issued a black-box warning (49). There are 2 subtypes of ARIA: ARIA-E (edema/effusion) and ARIA-H (hemorrhage). Though most ARIA cases are clinically asymptomatic, fatalities have been reported (45,50). ARIA occurs in 10%–30% of patients treated with anti-amyloid immunotherapy (46,47); thus, MRI monitoring is essential to screen for ARIA at baseline and throughout therapy (44,51,52). More than 4 baseline cerebral microhemorrhages, any site of superficial siderosis, and significant vascular disease are considered contraindications to treatment with anti-amyloid immunotherapy (53). ARIA rates are significantly higher in patients homozygous for the *APOE*-ε4 allele (54,55).

Approval for monoclonal anti-amyloid therapy is limited to patients with mild cognitive impairment or early AD, as patients with advanced-stage disease show limited benefit (56). To qualify for treatment with lecanemab or

donanemab therapy, biomarker confirmation of amyloid positivity must be established via PET or CSF biomarkers (51).

Amyloid PET in Therapy Monitoring

In addition to the diagnosis of AD, amyloid PET has been instrumental in follow-up imaging to monitor target engagement and response to therapy (Fig. 4). Serial amyloid PET imaging helps determine the extent of plaque reduction and guides treatment decisions (12). The FDA-approved labeling for both lecanemab and donanemab do not specify the exact timeline of treatment; early trial results covered 18 mo of treatment. The labeling for donanemab recommends considering stopping administration on the basis of the reduction of plaques to “minimal levels” or predefined thresholds (46,47). Some centers, including the Mayo Clinic, perform individual follow-up amyloid

PET scans at approximately 12 mo to assess the degree of plaque clearance and help determine whether to continue treatment. For the donanemab trials, therapy was stopped once amyloid levels fell below 11 Centiloids, indicating successful clearance, or if the Centiloid value was less than 24 for more than 2 follow-up scans (48). This approach allows for individualized treatment duration, limiting unnecessary drug exposure after treatment-related amyloid clearance is achieved (34). Full treatment-related amyloid clearance (Fig. 4) indicates that a patient’s PET-detected Aβ levels fell below a predetermined positivity threshold after treatment.

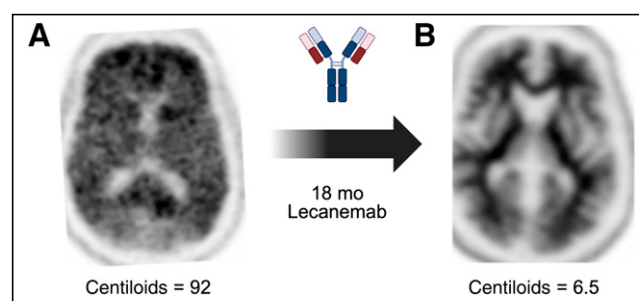


FIGURE 4. Pretreatment and posttreatment amyloid PET for anti-amyloid treatment monitoring. This patient had visually positive ^{11}C -Pittsburgh compound B amyloid PET scan, with concordant Centiloid value of 92. After 18 mo of anti-amyloid therapy, this patient’s florbetapir amyloid PET scan is no longer visually positive (6.5 Centiloids) and was reported as visually negative with complete amyloid clearance. Amyloid PET has advantage of visualizing these changes over other biomarkers. Patient and their neurologist opted to end therapy given amyloid clearance. Figure created with BioRender.

QUANTIFICATION AND INTERPRETATION OF AMYLOID PET IMAGING

SUV Ratio (SUVR) and the Centiloid Framework

To ensure accurate comparison of amyloid burden across patients and imaging centers, quantitative interpretation of amyloid PET has been standardized using metrics such as the SUVR and Centiloid scale. SUVR represents the mean radiopharmaceutical uptake within cortical regions of interest relative to an amyloid-negative reference region, typically the cerebellum (57). Because SUVR can vary across different amyloid PET radiopharmaceuticals, scanners, and analysis methods, the Centiloid scale was developed. This standardized linear transformation allows direct comparison of amyloid PET results obtained using different imaging protocols and radiopharmaceuticals (58). The Centiloid scale is an unbounded linear scale: 0 Centiloids represents mean uptake in young amyloid-negative control subjects, whereas 100 Centiloids represents typical uptake in patients with typical mild-to-moderate AD. Centiloid values may fall below 0 or above 100, depending on amyloid burden and the method used to convert raw SUVRs into Centiloids. Clinically, Centiloid values above 15–30 are typically considered indicative of amyloid positivity (59). However, quantitative results should always be interpreted in conjunction with the visual read, scan quality, and clinical context.

Standard quantification of amyloid PET is becoming increasingly important as more physicians use this imaging modality to assess therapeutic response to monoclonal anti-amyloid agents (60). As mentioned previously, amyloid PET burden guided therapy duration in the donanemab clinical trials. Patients with Centiloid values of 11 or less were

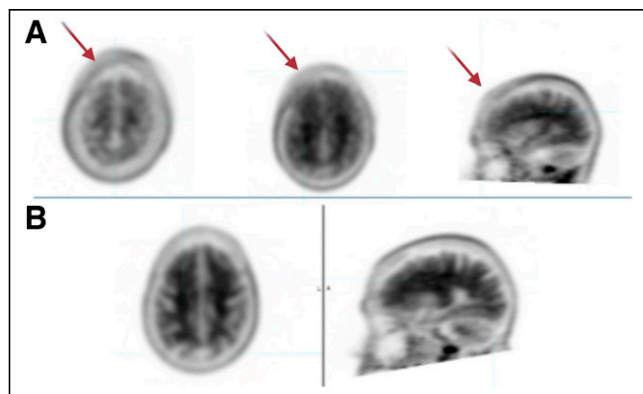


FIGURE 5. Automated quantification software can produce misleading results in presence of motion artifacts. These are images of same patient on 2 different days of florbetapir imaging. (A) motion artifacts, particularly in frontal bone and lobes (red arrows), can be seen. Centiloid value on this image was quantified at 77. (B) This patient returned to department for repeat imaging with head stabilization, and Centiloid value was quantified at 65. Although repeat imaging did not change positive visual read, change in quantification could be important for future follow-up comparisons if this patient receives anti-amyloid antibody therapy. Figure created with BioRender.

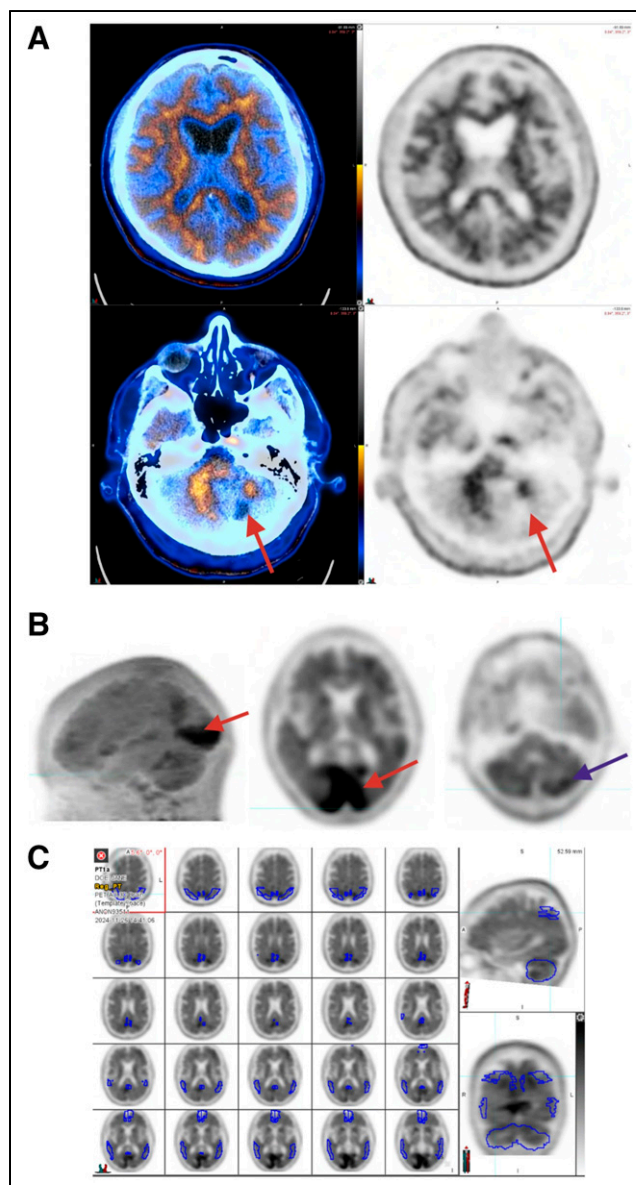


FIGURE 6. Automated quantification software can produce misleading results in presence of cerebellar artifacts. (A) Coronal amyloid PET scan depicting decreased left cerebellar uptake in encephalomalacia from prior infarct (arrows), which artificially inflated SUVR/Centiloid values. This patient's Centiloid value was calculated at 46, but this scan is visually negative. (B and C) Maximum-intensity-projection, sagittal, and coronal florbetapir images depicting increased cerebellar intake (purple arrow), which may artificially deflate SUVR/Centiloid values. This patient's Centiloid value was calculated at 17, but this examination was deemed visually positive. Note that intense occipital uptake (red arrows) is not included in automated region of interests applied for Centiloid calculation in C. Figure created with BioRender.

switched to placebo, those with Centiloid values of 11–25 received half-dose therapy before eventual discontinuation after 2 stable scans, and those with Centiloid values of more than 25 were continued on full-dose therapy until one of the clearance thresholds was reached (61).

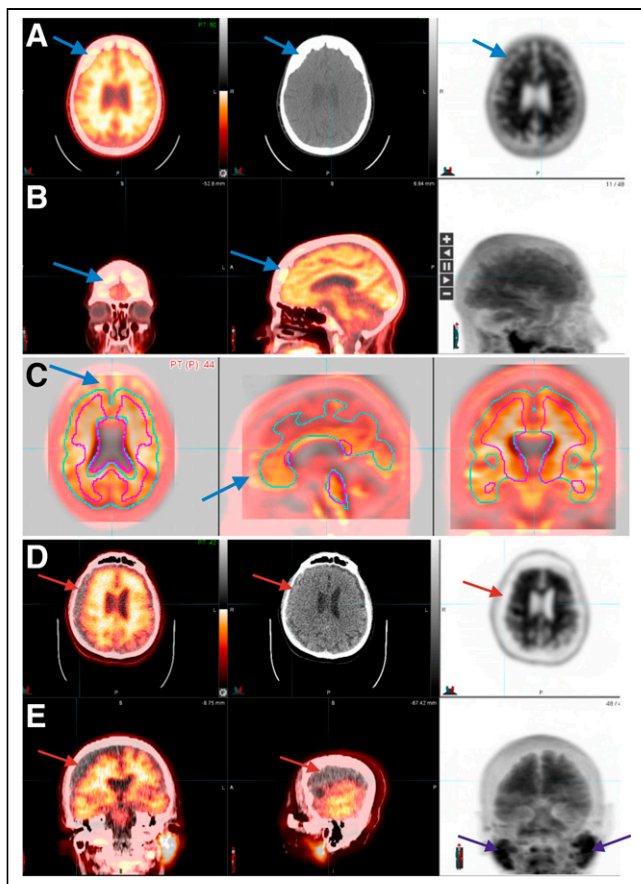


FIGURE 7. Examples of artifacts due to unrelated pathology or incidental findings on florbetapir amyloid PET. Axial (A) and sagittal (B) amyloid PET scans, with corresponding CT of patient with thickened skull secondary to hyperostosis frontalis interna (blue arrows). Note that there is amyloid radiopharmaceutical uptake in this bony process. It is imperative that, during quantitative analysis, regions of interest do not include areas of uptake outside of brain. (C) Correct region-of-interest placement. (D and E) Another patient had subdural hematoma (red arrows) on this amyloid PET scan. Patient was sent to emergency room; subdural hematoma was confirmed with diagnostic CT. Intense uptake in parotid and salivary glands (E, purple arrows) is nonspecific and does not necessitate follow-up imaging or treatment. Figure created with BioRender.

Although SUVR and the Centiloid scale provide valuable objective measures of amyloid burden, technical and biologic factors may limit their accuracy. Automated quantification software can produce misleading results in the presence of motion artifacts (Fig. 5), atrophy, or misregistration (48). Important pitfalls include cortical atrophy, which may lead to false negatives, especially when anatomic imaging is not available (48); reference region artifacts, including decreased cerebellar uptake (Fig. 6A), that would artificially elevate SUVR/Centiloid values (49); elevated cerebellar reference region uptake, which can result in misleadingly low SUVR/Centiloid values (62) (Figs. 6B and 6C); and off-target radiopharmaceutical uptake or inaccurate reference region placement (Figs. 7A–7C) (50).

Of note, incidental findings and unrelated pathologies may impact region-of-interest placement or clinical management (Figs. 7D and 7E).

Common Pitfalls for Amyloid PET Image Interpretation

For these reasons, visual quality control of PET images remains essential. Experienced readers should inspect alignment of gray and white matter masks, confirm region-of-interest placement, and correlate the images with structural brain MRI or CT images, when available (63). Automated analysis should supplement, not replace, expert visual interpretation.

TAU PET: IMAGING DISEASE PROGRESSION

Tau PET radiopharmaceuticals bind preferentially to paired helical filament tau neurofibrillary tangles, which correlates closely with clinical severity and cognitive decline in AD (64). Although tau PET can detect tau in some other tauopathies, its clinical use is specifically approved for suspected AD. Whereas amyloid PET is often considered binary, tau PET provides a more nuanced, graded, regional picture of disease stage and severity, as tau deposition tends to follow a characteristic progression that mirrors clinical decline. In AD, tau accumulation typically begins in the entorhinal cortex and hippocampus, with later spread to the neocortex (65). Centralized quantification frameworks for tau PET, including CenTauR and Uni(τ), are being developed (66,67).

Because of its increasing diagnostic and prognostic value, tau PET has recently moved closer to clinical implementation. The FDA approved the first tau PET radiopharmaceutical (flortaucipir) in 2020 (68), and, as of 2024, CMS can reimburse these scans under new Hospital Outpatient Prospective Payment System criteria. Of note, medial temporal lobe flortaucipir accumulation, thought to be seen in the early stages of AD, is not considered “positive” per the FDA-approved labeling (18), although many interpreting clinicians do include this finding in their report. Second-generation tau PET tracers exhibit reduced off-target intraparenchymal signal but are not yet FDA approved or clinically available (69). In 2024, the American College of Radiology recognized the utility of tau PET for diagnosing atypical AD, determining patient eligibility for anti-amyloid monoclonal antibody therapy, and informing the prognosis of AD (25).

Flortaucipir is effective in imaging 3R and 4R AD tau aggregates but lacks adequate sensitivity and specificity for 3R or 4R tauopathies. Ongoing work to develop tauopathy-specific PET ligands may enable tauopathy-specific PET imaging (70). Of note, the donanemab trials showed that anti-A β immunotherapies had greater benefit in individuals with medium or low tau pathology (46,48) on tau PET. Preliminary evidence suggests that anti-A β antibodies may slow, but not reverse, tau accumulation (71).

In summary, tau PET is increasingly used in clinical practice to guide treatment planning and biologically stage AD,

and its use will likely continue to increase. In the future, tau PET could help determine which patients are most likely to benefit from intervention and when to initiate treatment.

CONCLUSION

Molecular neuroimaging has revolutionized our understanding and management of AD. It has transformed the field from reliance on phenotypical, symptom-based diagnosis with postmortem confirmation to in vivo visualization of underlying pathology and a biomarker-driven framework for diagnosis and treatment. Amyloid, tau, and ^{18}F -FDG PET enable increasingly accurate diagnosis, staging, and therapeutic monitoring.

In the United States, the convergence of successful outcomes in clinical trials, FDA approval, CMS reimbursement, and the availability of new, effective therapies has ushered in a new era of personalized neuroimaging in AD. Challenges remain, including standardization across radio-pharmaceuticals, disparities in access, and education of clinicians and patients. However, PET imaging has become central to the diagnosis, biologic definition, disease staging, and ongoing management of AD. Leveraging PET, MRI, plasma, and CSF biomarker data, in conjunction with clinical and demographic information, stands to transform how clinicians understand and treat AD.

DISCLOSURE

No potential conflict of interest relevant to this article was reported.

REFERENCES

- 2025 Alzheimer's disease facts and figures. *Alzheimers Dement*. 2025;21:e70235.
- Braak H, Braak E. Diagnostic criteria for neuropathologic assessment of Alzheimer's disease. *Neurobiol Aging*. 1997;18(suppl):S85–S88.
- Hansson O. Biomarkers for neurodegenerative diseases. *Nat Med*. 2021;27:954–963.
- Franklin EE, Perrin RJ, Vincent B, et al. Brain collection, standardized neuropathologic assessment, and comorbidity in Alzheimer's Disease Neuroimaging Initiative 2 participants. *Alzheimers Dement*. 2015;11:815–822.
- Schneider JA, Arvanitakis Z, Leurgans SE, Bennett DA. The neuropathology of probable Alzheimer disease and mild cognitive impairment. *Ann Neurol*. 2009;66:200–208.
- Nelson PT, Jicha GA, Schmitt FA, et al. Clinicopathologic correlations in a large Alzheimer disease center autopsy cohort: neuritic plaques and neurofibrillary tangles “do count” when staging disease severity. *J Neuropathol Exp Neurol*. 2007;66:1136–1146.
- Bloom GS. Amyloid- β and tau: the trigger and bullet in Alzheimer disease pathogenesis. *JAMA Neurol*. 2014;71:505–508.
- Martin JB. Molecular basis of the neurodegenerative disorders. *N Engl J Med*. 1999;340:1970–1980.
- Kepp KP, Robakis NK, Hoiland-Carlsen PF, Sensi SL, Vissel B. The amyloid cascade hypothesis: an updated critical review. *Brain*. 2023;146:3969–3990.
- Frisoni GB, Hansson O, Nichols E, et al. New landscape of the diagnosis of Alzheimer's disease. *Lancet*. 2025;406:1389–1407.
- Minoshima S, Cross D, Thientunyakit T, Foster NL, Drzezga A. F-FDG PET imaging in neurodegenerative dementing disorders: insights into subtype classification, emerging disease categories, and mixed dementia with copathologies. *J Nucl Med*. 2022;63(suppl 1):2S–12S.
- Chapleau M, Iaccarino L, Soleimani-Meigooni D, Rabinovici GD. The role of amyloid PET in imaging neurodegenerative disorders: a review. *J Nucl Med*. 2022;63(suppl 1):13S–19S.
- Ivanidze J, Nordvig AS, Fajardo AR, Tsiouris AJ, Chiang GC, Osborne JR. Medicare coverage of amyloid PET: implications for clinical practice. *AJNR*. 2023;44:E45.
- Jack CR, Jr, Andrews JS, Beach TG, et al. Revised criteria for diagnosis and staging of Alzheimer's disease: Alzheimer's Association Workgroup. *Alzheimers Dement*. 2024;20:5143–5169.
- Benson DF, Kuhl DE, Hawkins RA, Phelps ME, Cummings JL, Tsai SY. The fluorodeoxyglucose ^{18}F scan in Alzheimer's disease and multi-infarct dementia. *Arch Neurol*. 1983;40:711–714.
- Minoshima S, Giordani B, Berent S, Frey KA, Foster NL, Kuhl DE. Metabolic reduction in the posterior cingulate cortex in very early Alzheimer's disease. *Ann Neurol*. 1997;42:85–94.
- Cohen AD, Villemagne VL. A brief history of A β imaging. *Alzheimers Dement*. 2025;21:e70291.
- Fleisher AS, Pontecorvo MJ, Devous MD, Sr, et al.; A16 Study Investigators. Positron emission tomography imaging with [^{18}F]flortaucipir and postmortem assessment of Alzheimer disease neuropathologic changes. *JAMA Neurol*. 2020;77:829–839.
- FDG PET for dementia and neurodegenerative diseases. Centers for Medicare and Medicaid Services website. <https://www.cms.gov/medicare-coverage-database/view/ncd.aspx?NCDId=288&ncdver=3&>. Published April 3, 2009. Accessed January 22, 2026.
- Xiang X, Wind K, Wiedemann T, et al. Microglial activation states drive glucose uptake and FDG-PET alterations in neurodegenerative diseases. *Sci Transl Med*. 2021;13:eabe5640.
- Zimmer ER, Parent MJ, Souza DG, et al. [^{18}F]FDG PET signal is driven by astroglial glutamate transport. *Nat Neurosci*. 2017;20:393–395.
- Marcus C, Mena E, Subramaniam RM. Brain PET in the diagnosis of Alzheimer's disease. *Clin Nucl Med*. 2014;39:e413–e422.
- Minoshima S, Foster NL, Sima AA, Frey KA, Albin RL, Kuhl DE. Alzheimer's disease versus dementia with Lewy bodies: cerebral metabolic distinction with autopsy confirmation. *Ann Neurol*. 2001;50:358–365.
- Kantarci K, Boeve BF, Przybelski SA, et al. FDG PET metabolic signatures distinguishing prodromal DLB and prodromal AD. *Neuroimage Clin*. 2021;31:102754.
- Soderlund KA, Austin MJ, Ben-Haim S, et al; Expert Panel On Neurological Imaging. ACR Appropriateness Criteria® dementia: 2024 update. *J Am Coll Radiol*. 2025;22:S202–S233.
- Perovnik M, Vo A, Nguyen N, et al. Automated differential diagnosis of dementia syndromes using FDG PET and machine learning. *Front Aging Neurosci*. 2022;14:1005731.
- Barnard L, Botha H, Corriveau-Lecavalier N, et al. An FDG-PET-based machine learning framework to support neurologic decision-making in Alzheimer disease and related disorders. *Neurology*. 2025;105:e213831.
- Rabinovici GD, Knopman DS, Arbizu J, et al. Updated appropriate use criteria for amyloid and tau PET: a report from the Alzheimer's Association and Society for Nuclear Medicine and Molecular Imaging Workgroup. *Alzheimers Dement*. 2025;21:e14338.
- Yang L, Rieves D, Ganley C. Brain amyloid imaging—FDA approval of florbetapir F18 injection. *N Engl J Med*. 2012;367:885–887.
- Lesman-Segev OH, La Joie R, Iaccarino L, et al. Diagnostic accuracy of amyloid versus ^{18}F -fluorodeoxyglucose positron emission tomography in autopsy-confirmed dementia. *Ann Neurol*. 2021;89:389–401.
- FDA grants supplemental indication approvals to three amyloid PET imaging agents, expanding the role of nuclear medicine in neurology. Society of Nuclear Medicine and Molecular Imaging website. <https://snmmi.org/Web/News/Articles/FDA-Grants-Supplemental-Indication-Approvals-to-Three-Amyloid-PET-Imaging-Agents-Expanding-the-Role.aspx>. Accessed December 20, 2025.
- Ackley SF, Wang J, Chen R, Power MC, Allen IE, Glymour MM. Estimated effects of amyloid reduction on cognitive change: a Bayesian update across a range of priors. *Alzheimers Dement*. 2024;20:1149–1155.
- Mila-Aloma M, Tosun D, Schindler SE, et al. Timing of changes in Alzheimer's disease plasma biomarkers as assessed by amyloid and tau PET clocks. *Ann Neurol*. 2025;98:508–523.
- La Joie R, Cummings JL, Dage JL, et al. Treatment-related amyloid clearance (TRAC): a framework to characterize patients in the era of anti-amyloid therapies. *Alzheimers Dement*. 2025;21:e70997.
- Piura YD, Figdore DJ, Lachner C, et al. Diagnostic performance of plasma p-tau217 and A β 42/40 biomarkers in the outpatient memory clinic. *Alzheimers Dement*. 2025;21:e70316.
- Sexton ZA, Perl JR, Saul HR, et al. Time from authorization by the US Food and Drug Administration to Medicare coverage for novel technologies. *JAMA Health Forum*. 2023;4:e232260.
- Rabinovici GD, Gatsonis C, Apgar C, et al. Association of amyloid positron emission tomography with subsequent change in clinical management among Medicare beneficiaries with mild cognitive impairment or dementia. *JAMA*. 2019;321:1286–1294.

38. Beta amyloid positron emission tomography in dementia and neurodegenerative disease. Centers for Medicare and Medicaid Services website. <https://www.cms.gov/medicare-coverage-database/view/ncacal-decision-memo.aspx?proposed=N&ncid=308>. Published October 13, 2023. Accessed December 22, 2025.
39. Hoy SM. Lecanemab: first approval. *Drugs*. 2023;83:359–365.
40. Statement: broader Medicare coverage of Leqembi available following FDA traditional approval. Centers for Medicare & Medicaid Services website. <https://www.cms.gov/newsroom/press-releases/statement-broader-medicare-coverage-leqembi-available-following-fda-traditional-approval>. Published July 6, 2023. Accessed December 22, 2025.
41. Windon CC, Gatsonis C, Carrillo MC, et al. Amyloid PET and clinical management in a diverse, cognitively impaired population: the new IDEAS study. *Alzheimers Dement*. 2025;21:e70504.
42. Heidebrink JL, Paulson HL. Lessons learned from approval of aducanumab for Alzheimer's disease. *Annu Rev Med*. 2024;75:99–111.
43. Kang C. Donanemab: first approval. *Drugs*. 2024;84:1313–1318.
44. Cummings J, Apostolova L, Rabinovici GD, et al. Lecanemab: appropriate use recommendations. *J Prev Alzheimers Dis*. 2023;10:362–377.
45. Zimmer JA, Ardayfio P, Wang H, et al. Amyloid-related imaging abnormalities with donanemab in early symptomatic Alzheimer disease: secondary analysis of the TRAILBLAZER-ALZ and ALZ 2 randomized clinical trials. *JAMA Neurol*. 2025;82:461–469.
46. Sims JR, Zimmer JA, Evans CD, et al.; TRAILBLAZER-ALZ 2 Investigators. Donanemab in early symptomatic Alzheimer disease: the TRAILBLAZER-ALZ 2 randomized clinical trial. *JAMA*. 2023;330:512–527.
47. van Dyck CH, Swanson CJ, Aisen P, et al. Lecanemab in early Alzheimer's disease. *N Engl J Med*. 2023;388:9–21.
48. Mintun MA, Lo AC, Duggan Evans C, et al. Donanemab in early Alzheimer's disease. *N Engl J Med*. 2021;384:1691–1704.
49. Agarwal A, Gupta V, Brahmabhatt P, et al. Amyloid-related imaging abnormalities in Alzheimer disease treated with anti-amyloid- β therapy. *Radiographics*. 2023;43:e230009.
50. Solopova E, Romero-Fernandez W, Harmsen H, et al. Fatal iatrogenic cerebral β -amyloid-related arteritis in a woman treated with lecanemab for Alzheimer's disease. *Nat Commun*. 2023;14:8220.
51. Rabinovici GD, Selkoe DJ, Schindler SE, et al. Donanemab: appropriate use recommendations. *J Prev Alzheimers Dis*. 2025;12:100150.
52. FDA to recommend additional, earlier MRI monitoring for patients with Alzheimer's disease taking Leqembi (lecanemab). Food and Drug Administration website. <https://www.fda.gov/drugs/drug-safety-and-availability/fda-recommend-additional-earlier-mri-monitoring-patients-alzheimers-disease-taking-leqembi-lecanemab>. Published August 8, 2025. Accessed December 26, 2025.
53. Withington CG, Turner RS. Amyloid-related imaging abnormalities with anti-amyloid antibodies for the treatment of dementia due to Alzheimer's disease. *Front Neurol*. 2022;13:862369.
54. Weber M, Roth HL, Abramowitz A, Johnson KG. Developing treatment models for the delivery of the anti-amyloid therapy, lecanemab: considerations for implementation of lecanemab in healthcare systems. *Am J Geriatr Psychiatry*. 2025;33:601–610.
55. Teipel S, Tang Y, Khachaturian A. Clinical efficacy of anti-amyloid antibodies in apolipoprotein E ϵ 4 homozygotes: a Bayesian reanalysis of lecanemab and donanemab phase 3 results. *Alzheimers Dement (NY)*. 2025;11:e70083.
56. Jucker M, Walker LC. Alzheimer's disease: from immunotherapy to immunoprevention. *Cell*. 2023;186:4260–4270.
57. Minoshima S, Drzezga AE, Barthel H, et al. SNMMI procedure standard/EANM practice guideline for amyloid PET imaging of the brain 1.0. *J Nucl Med*. 2016;57:1316–1322.
58. Klunk WE, Koeppe RA, Price JC, et al. The Centiloid Project: standardizing quantitative amyloid plaque estimation by PET. *Alzheimers Dement*. 2015;11:1–15.e1–14.
59. Iaccarino L, Burnham SC, Tunali I, et al. A practical overview of the use of amyloid-PET Centiloid values in clinical trials and research. *Neuroimage Clin*. 2025;46:103765.
60. Pemberton HG, Collij LE, Heeman F, et al.; AMYPAD Consortium. Quantification of amyloid PET for future clinical use: a state-of-the-art review. *Eur J Nucl Med Mol Imaging*. 2022;49:3508–3528.
61. Gueorguieva I, Willis BA, Chua L, et al. Donanemab population pharmacokinetics, amyloid plaque reduction, and safety in participants with Alzheimer's disease. *Clin Pharmacol Ther*. 2023;113:1258–1267.
62. Uneus EI, Wilhelmsson C, Backstrom D, et al. Cerebellar and cerebral amyloid visualized by [18 F]flutemetamol PET in long-term hereditary V30M (p.V50M) transthyretin amyloidosis survivors. *Front Neurol*. 2022;13:816636.
63. Tanabe J, Lim MF, Dash S, et al. Automated volumetric software in dementia: help or hindrance to the neuroradiologist? *AJNR*. 2024;45:1737–1744.
64. Scheltens P, De Strooper B, Kivipelto M, et al. Alzheimer's disease. *Lancet*. 2021;397:1577–1590.
65. Ossenkoppele R, Schonhaut DR, Scholl M, et al. Tau PET patterns mirror clinical and neuroanatomical variability in Alzheimer's disease. *Brain*. 2016;139:1551–1567.
66. Villemagne VL, Leuzy A, Bohorquez SS, et al. CenTauR: toward a universal scale and masks for standardizing tau imaging studies. *Alzheimers Dement (Amst)*. 2023;15:e12454.
67. Povala GB, Bellaver B, Lussier FZ, et al. Universal tau PET scale (Unit)—the HEAD study. *Alzheimers Dement*. 2025;20:e094383.
68. Franceschi AM, Keir G, Benzinger TLS, et al.; American Society of Functional Neuroradiology. Clinical role of brain PET in Alzheimer disease in the era of disease-modifying therapies. *AJNR*. 2025;46:1521–1527.
69. Bischof GN, Dodich A, Boccardi M, et al. Clinical validity of second-generation tau PET tracers as biomarkers for Alzheimer's disease in the context of a structured 5-phase development framework. *Eur J Nucl Med Mol Imaging*. 2021;48:2110–2120.
70. Liang SH. Development of novel PET tracers for imaging non-AD tauopathies. *ACS Med Chem Lett*. 2025;16:1884–1888.
71. Shcherbinin S, Evans CD, Lu M, et al. Association of amyloid reduction after donanemab treatment with tau pathology and clinical outcomes: the TRAILBLAZER-ALZ randomized clinical trial. *JAMA Neurol*. 2022;79:1015–1024.